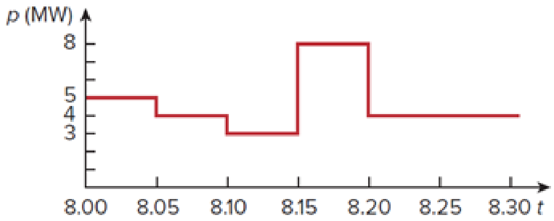
Chapter 1, Problem 35CP

Problem

The graph in Fig. 1.33 represents the power drawn by an industrial plant between 8:00 and 8:30 a.m. Calculate the total energy in MWh consumed by the plant.

Figure 1.33:



Step-by-step solution

Step 1 of 1

Total Energy between is

Power (mW) Time (Minutes)

5 5

4 5

3 5

8 5

4 10

$$\text{Total Energy} = \left(5\times\frac{5}{60}\right)+\left(4\times\frac{5}{60}\right)+\left(3\times\frac{5}{60}\right)+\left(8\times\frac{5}{60}\right)+\left(4\times\frac{10}{60}\right)$$

$$= \frac{14}{6}$$

$$= 2.33 \text{ mWh}$$